

Nawal Shahid

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SUMMARY

Full stack engineer with hands-on experience owning production features end to end, from REST API design and role-based authentication to React frontends and MongoDB performance tuning. Delivered a multi-role manufacturing platform, a MERN ed-tech product used across web and mobile, and two AI applications using Groq, FAISS, RAG, and prompt engineering. Strong in JavaScript, TypeScript, Python, Node.js, Express.js, React, SQL and NoSQL database design, and CI/CD.

EXPERIENCE

Full Stack Engineer, Production Management System

2026

Textile Manufacturing Client, Tanzania (Remote Contract)

- Architected a multi-role production management system covering inventory tracking, order processing, production workflows, and financial reporting, built on a Node.js and Express.js REST API with JWT authentication and role-based authorization across Admin, Manager, and Operator roles.
- Designed MongoDB document schemas with embedded references and compound indexes, improving query performance for reporting and inventory lookups across high-volume collections.
- Automated end-to-end business workflows, including procurement approvals, low-stock alerts, and report generation, and deployed to production with environment-based configuration and structured error handling.
- Standardized input validation and RESTful endpoint design across inventory, production, procurement, and reporting modules, reducing integration effort for future feature development.

Full Stack Web and Mobile Developer, DigiLex Ed-Tech Platform

2024 - 2026

Kazim Trust, Karachi, Pakistan

- Architected the complete MERN stack platform, including a Node.js and Express.js REST API with JWT authentication, role-based access control, and MongoDB schemas with compound indexes for efficient content retrieval and user progress queries.
- Engineered the React.js frontend with semantic HTML, keyboard navigation, and ARIA labeling, achieving a Lighthouse accessibility score of 93.
- Built a cross-platform React Native mobile app with Firebase Authentication and Firestore offline persistence, reducing session drop-off by approximately 40 percent.
- Improved application performance for low-bandwidth users through lazy loading, route-based code splitting, offline persistence, and reusable React component architecture.

Web Developer Apprentice, AJD Marketing

2023 - 2024

Qatar (Remote)

- Delivered and maintained 4 client websites using React.js, Node.js, PHP, MySQL, and Tailwind CSS, raising Lighthouse performance scores from 68 to 91 and cutting average page load time from 3.4 seconds to 1.7 seconds through Core Web Vitals tuning and SQL query optimization.

Open Source Contributor, GitHub

2026 -

Present

Remote, Various Repositories

- Resolved bugs and submitted pull requests across open source repositories, improving test coverage and documentation within large existing codebases.
- Collaborated with maintainers following established Git branching workflows and code review standards to maintain code quality.

PROJECTS

IntelliAI, AI-Powered Business Intelligence Platform

Streamlit, Groq Llama 3.3 70B, Plotly, Pandas, Prompt Engineering

- Engineered an end-to-end, no-code analytics platform with an 8-step pipeline covering CSV and Excel ingestion, automatic encoding and delimiter detection, date parsing, data profiling, cleaning, exploratory data analysis, and interactive Plotly dashboards.
- Integrated Groq's Llama 3.3 70B through prompt-engineered pipelines to generate business insights, detect anomalies, identify trends, and power conversational Q&A analytics and automated reporting.
- Designed a modular data-processing architecture separating preprocessing, AI-assisted analysis, and visualization, enabling non-technical business users to derive insights from raw data independently.

AI Study Brain, RAG Document Q&A Application

Python, Streamlit, FAISS, Sentence Transformers, Groq API

- Architected a modular Retrieval-Augmented Generation pipeline covering PDF upload, text extraction, configurable chunking, local sentence-transformer embeddings, FAISS vector indexing, and Groq API-generated answers with source citations.
- Designed the retrieval architecture to support semantic document search and context-aware question answering across multiple document types.

NexaPress, Full Stack CMS

MongoDB, Express.js, React.js, Node.js, JWT, Firebase, Cloud Functions, Vercel

- Built two parallel CMS architectures: a MERN version with JWT authentication, role-based access control, and dynamic API routing deployed on Vercel, and a serverless Firebase version with Firestore security rules, real-time sync, and Cloud Functions for automated publication workflows.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Kotlin

Frontend and Mobile: React.js, Next.js, HTML5, Tailwind CSS, Material UI, Semantic HTML, ARIA, Accessibility, React Native

Backend and APIs: Node.js, Express.js, REST API Design, JWT, Role-Based Access Control, Serverless, System Design, Debugging

Databases: MongoDB, Mongoose, Firestore, PostgreSQL, MySQL, Schema Design, Indexing, Aggregation Pipelines

AI Engineering: Groq LLM, Retrieval-Augmented Generation, FAISS, Sentence Transformers, Prompt Engineering, Pandas, Streamlit, Plotly

DevOps and Tools: Expo, Firebase, Docker, GitHub Actions, CI/CD, Git, Code Review, Agile, Netlify, Vercel

EDUCATION

Master of Science, Computer Science

2026 - 2028

Jinnah University for Women, Karachi, Pakistan

Bachelor of Science, Software Engineering

2022 - 2026

Jinnah University for Women, Karachi, Pakistan | CGPA: 3.5 / 4.0

- Technical Management Co-Lead, Google Developer Student Club (GDSC)
- Creative Art Manager, University Event Branding